Devin DeCosmo

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Education

Carnegie Mellon University *Pittsburgh, PA*
Master of Science in AI Engineering – Mechanical Engineering *December 2026*
Relevant Coursework: Systems and Toolchains for Engineers, AI/ML for Engineers
Arizona State University *Tempe, AZ*
Mechanical Engineering (BSE) *May 2025*
4.0/4.0 GPA, Summa Cum Laude, Barrett, The Honors College

Skills

Programming Languages: Python, C++, C#, MATLAB, SQL, Arduino, LabVIEW
ML & Data Science Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
Developer Tools: Git, Linux, Windows, Apache Spark, Neo4J, GCS
ML Techniques: Data Preprocessing, Data Augmentation, Transfer Learning, Computer Vision

Project Experience

Lanternfly Tracker: *Hackathon* *September 2025 - Present*

- Led ML and datalogging development for a webapp to crowdsource invasive lanternfly sightings, earning 2nd place out of 17 teams at HackCMU, demonstrating a scalable prototype for monitoring the range of invasive insects
- Collected and augmented 300+ real images of lanternflies and 2,000 insect/environmental images from Hugging Face to build a robust 5,000 image dataset
- Trained, and evaluated an EfficientNet-B0 classifier using Python and TensorFlow an EfficientNet-B0 classifier (**Python, TensorFlow**) using transfer learning, reaching 89.7% accuracy in identifying lanternfly concentrations with a lightweight, user-ready model
- Implemented real-time CSV data logging (**Numpy**) and graphing using kernel density mapping (**SciPy, Matplotlib**) to generate live concentration visualizations for users and researchers

Thread Checker: *ML Personal Project* *August 2025 - Present*

- Developed a computer vision program to classify different thread types on fasteners across 3 categories, creating a dataset of 400 real images augmented to 4,000 samples
- Implemented data preprocessing and data augmentation to standardize scaling, aspect ratio, and contrast, improving classification accuracy by 15%
- Trained a MobileNetV2 classifier (**Python, Keras, TensorFlow**) with custom pooling layers, achieving 76% testing accuracy across all thread types
- Deployed the model to a Hugging Face Space for live testing and demonstration

Stock Binary Sentiment Analysis Using 10-Q Forms: *Class Project* *October 2024 - December 2024*

- Developed a dual-model machine learning system to predict stock price movement post-earnings using **Tensorflow** with a FinBERT transformer and shallow neural network for sentiment analysis, achieving a peak classification accuracy of 75%
- Built an automated data pipeline utilizing the SEC's Python API and web scraping to successfully aggregate unstructured 10-Q text and structured financial data, enabling rapid organization of financial data for 20 publicly traded companies
- Conducted feature analysis with a FinBERT model by training on different report sections, improving qualitative sentiment analysis utilizing **MatplotLib** to identify that a combination of all text features yielded the highest validation accuracy (75%)
- Engineered and trained a shallow neural network using **TensorFlow** on key financial metrics (e.g., EPS, ROE) for rapid classification, achieving 70% testing accuracy

ASU Senior Capstone Project: *Dept of the Army* *August 2024 - May 2025*

- Led a 6-person team in designing a reusable modular target system for military exercises; gathered requirements, developed modular target geometries, and delivered documentation for assembly and maintenance 1 month ahead of schedule

Zigzag Flow Reactor: *ASU Barrett Honors Thesis* *January 2022 - May 2025*

- Redesigned mechanical and electrical systems for a laboratory reactor to characterize energy storage medium, CAM28, identifying faster reduction reactions than suggested by past research

Professional Experience

Engineering Intern: *RedoxBlox* *San Diego, CA: May 2025 - August 2025*

- Managed development of high-temperature control system for DOE project, concurrently executing mechanical design, control logic (**Arduino**), and GUI (**C#**) achieving ±1% control tolerance on an aggressive schedule
- Engineered and deployed a PID control system (**Arduino**), using **Git** within a **Linux environment** to manage development and deliver the project 1 week early
- Engineered a **C#** application for real-time data monitoring and system control, enabling live data streaming, visualization, and logging via an HTTP client

Mechanical Engineering Intern: *Dept. of the Army* *Tobyhanna PA: May 2024 - August 2024*

- Designed four custom transmitter components for an Air Force threat emitter, collaborating with technical and administrative staff to allow delivery 2 weeks ahead of schedule and $1,000 below budget